

NYC TAXI DEMAND PREDICTION

Personal Project Documentation

An End-to-End Data Analytics and Machine Learning Project

Domain: *Transportation Analytics*

Primary Tools: *Python, Pandas, NumPy, Matplotlib, Scikit-learn,
Jupyter Notebook, Power BI*

Machine Learning Model: *Random Forest Regression*

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1. PROJECT OVERVIEW

The **NYC Taxi Demand Prediction** project is an end-to-end personal data analytics and machine learning project developed to analyze taxi demand patterns and predict hourly taxi pickup demand across different pickup zones.

The project uses NYC Yellow Taxi trip data and follows a complete data analytics workflow, beginning with data loading and preprocessing, followed by exploratory data analysis, demand aggregation, feature engineering, machine learning model development, model evaluation, and Power BI dashboard creation.

The main objective of the project is to understand how taxi demand changes across different hours and pickup locations and to determine whether historical demand patterns can be used to predict future hourly demand.

The project combines three major areas:

- **Data Analytics** using Python
- **Machine Learning** using Random Forest Regression
- **Business Intelligence and Visualization** using Power BI

The final machine learning model achieved a test **MAE of 3.63**, **RMSE of 11.86**, and **R² of 0.9627**, demonstrating strong predictive performance on the unseen test dataset.

The final results were presented through an interactive Power BI executive dashboard containing demand KPIs, hourly demand analysis, top pick up zones, borough-level demand, actual versus predicted demand, and model performance.

2. INTRODUCTION

Transportation systems generate large amounts of data that can be analyzed to understand travel behavior and demand patterns.

Taxi trip data is particularly useful because it contains information about when and where taxi pickups occur. By analyzing historical taxi activity, it is possible to identify periods and locations with higher demand.

Taxi demand is not evenly distributed throughout the day. Demand can change depending on:

- Time of day

- Day of the week
- Pickup location
- Recent historical demand
- Previous-day demand patterns

Understanding these patterns can help provide a clearer picture of how taxi demand changes over time and across different locations.

This project focuses on NYC Yellow Taxi data and transforms individual taxi trip records into **hourly demand by pickup zone**.

The resulting dataset is then used to develop a machine learning model that predicts hourly taxi demand.

The project also uses Power BI to present the findings in a business-friendly format.

3. PROBLEM STATEMENT

Taxi demand varies significantly across different times and locations.

Without analyzing historical demand, it can be difficult to determine:

- Which hours experience the highest taxi demand
- Which pickup zones generate the most demand
- How demand differs between weekdays and weekends
- How previous demand affects future demand
- How accurately hourly taxi demand can be predicted

Therefore, the problem addressed in this project is:

How can NYC Yellow Taxi trip data be analyzed and transformed into hourly zone-level demand, and how can historical demand patterns be used to predict future taxi demand while presenting the results through an interactive dashboard?

4. PROJECT OBJECTIVES

The main objectives of this project were:

1. Analyze NYC taxi demand patterns.
2. Understand how demand changes throughout the day.
3. Identify high-demand pickup zones.
4. Analyze taxi demand across boroughs.
5. Compare weekday and weekend demand patterns.

6. Transform trip-level taxi data into hourly zone-level demand.
7. Create time-based features for demand analysis.
8. Create historical demand features for machine learning.
9. Develop a Random Forest Regression model for hourly demand prediction.
10. Use a chronological train-validation-test approach.
11. Evaluate the model using MAE, RMSE, and R^2 .
12. Compare the Random Forest model against a baseline.
13. Analyze feature importance.
14. Test model performance without the previous-hour demand feature.
15. Generate predictions for the unseen test dataset.
16. Build an interactive Power BI dashboard.
17. Communicate the analytical and machine learning results through visualizations and KPIs.

5. PROJECT SCOPE

The project covers the complete process from raw taxi trip data to an analytical and predictive dashboard.

5.1 Data Analytics

The analytics component includes:

- Data loading
- Data inspection
- Data cleaning
- Timestamp processing
- Demand aggregation
- Exploratory data analysis
- Hourly demand analysis
- Pickup zone analysis
- Borough-level analysis

5.2 Machine Learning

The machine learning component includes:

- Feature engineering

- Historical demand features
- Time-based dataset splitting
- Baseline evaluation
- Random Forest Regression
- Model validation
- Test evaluation
- Feature importance analysis
- Prediction generation

5.3 Business Intelligence

The Power BI component includes:

- KPI cards
- Hourly demand visualization
- Top pick up zone analysis
- Borough-level demand analysis
- Actual versus predicted demand
- Model performance summary

6. DATASET DESCRIPTION

The project uses **NYC Yellow Taxi trip data**.

The original data contains individual taxi trip records. For this project, the trip-level information was transformed into an hourly demand dataset based on pickup zones.

The transformation allowed the project to focus on the number of taxi pickups occurring within a particular pickup zone during a particular hour.

This created a structured dataset suitable for both demand analysis and machine learning.

6.1 Dataset Characteristics

The final modeling dataset contained:

Dataset Component	Number
Unique pickup zones	261
Training rows	144,072
Validation rows	18,792
Testing rows	25,056

The Power BI dashboard reports approximately:

3.47 million total taxi trips

for the analyzed period.

7. TOOLS AND TECHNOLOGIES

The following tools and technologies were used throughout the project.

7.1 Python

Python was used as the primary programming language for data preparation, analysis, feature engineering, machine learning, and evaluation.

7.2 Pandas

Pandas was used for:

- Loading datasets
- Data cleaning
- Data manipulation
- Grouping and aggregation
- Creating new columns
- Preparing the modeling dataset
- Working with prediction results

7.3 NumPy

NumPy was used for numerical operations and array-based processing.

7.4 Matplotlib

Matplotlib was used for data visualization during the analytical process.

7.5 Scikit-learn

Scikit-learn was used for:

- Random Forest Regression
- Model training
- Model evaluation
- Regression metrics
- Feature importance analysis

7.6 Jupyter Notebook

Jupyter Notebook was used to organize and execute the Python-based analysis and machine learning workflow.

7.7 Power BI

Power BI was used to create the final executive dashboard and present the analytical and predictive results.

7.8 CSV and Parquet

CSV and Parquet formats were used to store processed datasets, predictions, feature importance results, and model performance results.

8. PROJECT WORKFLOW

The complete project workflow can be summarized as follows:

NYC Yellow Taxi Trip Data



Data Loading



Data Inspection



Data Cleaning & Preprocessing



Exploratory Data Analysis



Hourly Zone-Level Demand Aggregation



Feature Engineering



Machine Learning Dataset Preparation



Time-Based Train / Validation / Test Split



Baseline Evaluation



Random Forest Regression



Model Evaluation



Feature Importance Analysis



Model Without lag_1



Test Predictions



Power BI Dashboard



Business and Technical Insights

9. DATA LOADING AND INITIAL INSPECTION

The first stage of the project involved loading the NYC Yellow Taxi trip data into the Python environment.

Before performing any analysis, the dataset was inspected to understand its structure and identify the fields required for the project.

The initial inspection focused on:

- Dataset structure
- Column names
- Data types
- Pickup timestamp information
- Pickup location information
- Potential missing or invalid records
- Variables required for demand aggregation

Understanding the structure of the dataset was necessary before performing data cleaning and transformation.

The pickup timestamp and pickup location information were particularly important because they were used to create hourly zone-level demand.

10. DATA CLEANING AND PREPROCESSING

Data cleaning and preprocessing were performed before exploratory analysis and machine learning.

The purpose of preprocessing was to ensure that the data was in a suitable format for demand aggregation and model development.

10.1 Loading the Data

The NYC Yellow Taxi data was loaded into Python for processing.

10.2 Inspecting the Dataset

The dataset was examined to understand its structure, data types, and available variables.

10.3 Handling Invalid or Unusable Records

Records that were not suitable for the analysis were handled during the preprocessing stage.

This helped reduce the effect of invalid observations on the subsequent analysis.

10.4 Converting Pickup Time to Datetime

The pickup timestamp was converted into datetime format.

This allowed different time components to be extracted, including:

- Date
- Hour
- Day of week

10.5 Extracting Time Components

The pickup timestamp was used to create time-related information needed for both analysis and modeling.

10.6 Identifying Pickup Zones

Pickup location information was retained to allow demand to be analyzed by zone.

The final dataset contained:

261 unique pickup zones.

10.7 Preparing the Data for Aggregation

The cleaned records were organized according to pickup time and pickup zone.

10.8 Creating Hourly Demand

Taxi trips were aggregated by hour and pickup zone.

This transformed individual trip records into hourly demand observations.

10.9 Preparing for Lag Features

The hourly demand dataset was ordered appropriately before creating historical demand features.

Lag and rolling calculations naturally produced missing values at the beginning of certain sequences. These affected rows were handled before model training.

11. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the major patterns in taxi demand before developing the machine learning model.

The analysis focused on:

- Demand by hour
- Demand by day type
- Demand by pickup zone
- Demand by borough
- High-demand locations
- Actual versus predicted demand

11.1 Hourly Demand Analysis

Hourly taxi demand was analyzed to understand how demand changes throughout the day.

The analysis showed that demand varies considerably across different hours.

Demand is relatively lower during early morning periods and increases during the daytime.

The Power BI dashboard identifies:

18:00

as the peak demand hour.

This indicates that the evening period represents an important high-demand period in the analyzed data.

11.2 Weekday and Weekend Analysis

Demand was analyzed based on whether the day was a weekday or weekend.

The `is_weekend` feature was used to distinguish between these two categories.

11.3 Pickup Zone Analysis

Taxi demand was analyzed across the 261 pickup zones.

The purpose was to identify locations with particularly high levels of taxi activity.

The Power BI dashboard presents the top five pickup zones.

The highest-demand pickup zone identified in the dashboard is:

Midtown Center

The dashboard also highlights other high-demand locations such as:

- Upper East Side South
- Upper East Side North
- JFK Airport
- Times Sq/Theatre District

11.4 Borough-Level Analysis

Demand was also analyzed at the borough level.

The Power BI dashboard displays:

- Manhattan
- Brooklyn
- Bronx
- EWR
- Staten Island

The analysis shows that a large share of the observed taxi demand is concentrated in Manhattan.

12. HOURLY ZONE-LEVEL DEMAND AGGREGATION

One of the most important transformations in the project was converting individual taxi trips into hourly demand by pickup zone.

Instead of treating every trip as a separate observation, the trips were grouped according to:

- Pickup zone
- Hour

The resulting dataset represents the number of taxi pickups associated with each pickup zone during each hour.

This transformation created the target variable required for the machine learning model.

For example, a particular pickup zone could have a demand value representing the number of taxi pickups recorded during a specific hour.

The resulting dataset was saved as:

hourly_zone_demand.parquet

The completed hourly demand dataset contained:

- **194,184 rows**
- **10 columns**
- **261 active zones**
- **0 missing demand values**

This confirmed that the hourly demand dataset had been successfully created before the modeling stage.

13. FEATURE ENGINEERING

Feature engineering was performed to create variables that could help the model identify relationships between time, location, historical demand, and future demand.

The final features used by the model were:

Feature	Description
PULocationID	Pickup zone identifier
hour	Hour of the day
day_of_week	Day of the week
day_of_month	Day of the month
is_weekend	Indicates whether the day is a weekend
lag_1	Demand from the previous hour
lag_24	Demand from the same hour on the previous day
rolling_24	Rolling 24-hour demand feature

13.1 PULocationID

PULocationID represents the pickup zone.

It allows the model to distinguish demand behavior between different pickup locations.

13.2 Hour

The hour feature represents the hour of the day.

This feature captures recurring daily demand patterns.

For example, demand during early morning hours may differ substantially from demand during evening hours.

13.3 Day of Week

The `day_of_week` feature represents the day of the week.

This allows the model to capture recurring weekly demand patterns.

13.4 Day of Month

The `day_of_month` feature represents the calendar day within the month.

It provides additional temporal information to the model.

13.5 Is Weekend

The `is_weekend` feature identifies whether an observation belongs to a weekend.

This allows the model to distinguish weekday and weekend behavior.

13.6 Lag 1

The `lag_1` feature represents demand from the previous hour.

This feature became the most important feature in the Random Forest model.

Its feature importance was:

0.894862

This indicates that recent demand provided a very strong predictive signal.

13.7 Lag 24

The `lag_24` feature represents demand from the same hour on the previous day.

Its feature importance was:

0.073075

This was the second most important feature in the model.

13.8 Rolling 24

The `rolling_24` feature represents a rolling 24-hour demand measure.

This feature provides information about recent demand behavior over a longer time window.

Its feature importance was:

0.008184

14. MACHINE LEARNING DATASET PREPARATION

After feature engineering, the data was prepared for machine learning.

The target variable represented hourly taxi demand.

The model input variables consisted of the engineered:

- Location features
- Time features
- Historical demand features
- Rolling demand information

Before model training, rows affected by unavailable lag and rolling values were removed where necessary.

The resulting dataset was then divided chronologically into training, validation, and test datasets.

15. TRAIN, VALIDATION AND TEST SPLIT

Because this project deals with time-dependent demand, a chronological split was used instead of randomly mixing the observations.

This approach helps maintain the temporal order of the data.

The final split was:

Dataset Rows

Training **144,072**

Validation **18,792**

Testing **25,056**

The training dataset was used to train the Random Forest model.

The validation dataset was used to evaluate model performance during development.

The test dataset was kept for the final evaluation on unseen observations.

This approach helped ensure that future observations were not used during model training.

16. BASELINE MODEL

A baseline was established before evaluating the Random Forest model.

The purpose of the baseline was to provide a reference point for evaluating whether the machine learning model provided an improvement.

The baseline performance was:

Metric Baseline

MAE **5.43**

RMSE **17.91**

R² **0.8938**

These values were later compared against the Random Forest test results.

17. RANDOM FOREST REGRESSION MODEL

A **Random Forest Regression** model was developed to predict hourly taxi demand.

Random Forest is an ensemble machine learning algorithm based on multiple decision trees.

Because the target variable in this project is numerical taxi demand, the regression version of Random Forest was used.

The model learned relationships between:

- Pickup zone
- Hour
- Day of week
- Day of month
- Weekend status
- Previous-hour demand
- Previous-day demand
- Rolling demand

The trained model was evaluated first on the validation dataset and then on the unseen test dataset.

18. MODEL EVALUATION METRICS

Three metrics were used to evaluate the regression model.

18.1 Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted demand.

A lower MAE indicates that the model's predictions are closer to the actual values.

The final test MAE was:

3.63

18.2 Root Mean Squared Error (RMSE)

RMSE measures the square root of the average squared prediction error.

Because the errors are squared before averaging, RMSE gives greater influence to larger prediction errors.

The final test RMSE was:

11.86

18.3 R² Score

R² measures how much of the variation in the target variable is explained by the model.

The final test R² was:

0.9627

or:

96.27%

Therefore, the model explained approximately **96.27% of the variance in the unseen test demand data.**

19. MODEL PERFORMANCE

The Random Forest model was first evaluated using the validation dataset.

19.1 Validation Performance

Metric	Result
MAE	4.17
RMSE	13.45
R ²	0.9402

The model was then evaluated on the unseen test dataset.

19.2 Final Test Performance

Metric	Result
MAE	3.63
RMSE	11.86
R ²	0.9627

The final test performance indicates that the model performed strongly on unseen data.

20. BASELINE VS RANDOM FOREST COMPARISON

The Random Forest model was compared against the baseline.

Metric	Baseline	Random Forest
MAE	5.43	3.63
RMSE	17.91	11.86
R ²	0.8938	0.9627

The Random Forest model achieved:

- Lower MAE
- Lower RMSE
- Higher R²

than the baseline.

This indicates that the Random Forest model provided better predictive performance than the baseline.

21. FEATURE IMPORTANCE ANALYSIS

Feature importance was examined to understand which variables contributed most strongly to the Random Forest predictions.

The results were:

Feature	Importance
lag_1	0.894862
lag_24	0.073075
hour	0.011241
rolling_24	0.008184
day of week	0.005440
PULocationID	0.003832
day of month	0.002943
is_weekend	0.000423

The results show that lag_1 was by far the most influential feature.

The second most influential feature was lag_24.

This suggests that recent historical demand was particularly useful for predicting the next hourly demand level.

22. MODEL WITHOUT lag_1

Because lag_1 was the dominant feature, an additional model was trained without it.

The purpose of this experiment was to determine whether the model could still perform well using the remaining features.

The features used in this experiment were:

PULocationID

hour

day_of_week

day_of_month

is_weekend

lag_24

rolling_24

Performance Without lag_1

Metric	Result
MAE	3.98
RMSE	12.71
R ²	0.9572

The model still achieved an R² of **0.9572**, or approximately **95.72%**.

However, performance decreased compared with the full model.

Metric	Full Model	Without lag_1
MAE	3.63	3.98
RMSE	11.86	12.71
R ²	0.9627	0.9572

This experiment confirmed that lag_1 provided a significant predictive contribution.

23. TEST PREDICTIONS

After training and evaluating the model, predictions were generated for the unseen test dataset.

The final test dataset contained:

25,056 observations

and therefore:

25,056 predictions

were generated.

The prediction results included:

- Datetime
- Pickup zone
- Actual demand
- Predicted demand

The prediction results were stored in:

ml_predictions.csv

These predictions were subsequently used in the Power BI dashboard to create the Actual vs Predicted Taxi Demand visualization.

24. POWER BI DASHBOARD

A Power BI executive dashboard was created to communicate the project's analytical and machine learning results.

The dashboard provides a single-page view of:

- Overall taxi demand
- Demand by hour
- Top pickup zones
- Borough-level demand
- Actual versus predicted demand
- Model performance

The dashboard was designed to allow a user to understand the main results quickly without having to examine the Python notebook.

Final Power BI Executive Dashboard:



25. DASHBOARD KPIS

Five main KPI cards were included in the dashboard.

25.1 Total Trips

3.47M

This represents the total number of taxi trips reported in the dashboard for the analyzed period.

25.2 Active Zones

261

This represents the number of pickup zones included in the analysis.

25.3 Peak Demand Hour

18:00

The dashboard identifies 18:00 as the peak demand hour.

25.4 Top Pickup Zone

Midtown Center

This is the highest-demand pickup zone identified in the dashboard.

25.5 Model R²

96.27%

This represents the final test R² of the Random Forest model.

26. DASHBOARD VISUALIZATIONS

The dashboard contains several visualizations designed to communicate both business and machine learning insights.

26.1 Hourly Taxi Demand

The hourly demand visualization shows how taxi demand changes throughout the day.

The chart compares weekday and weekend demand patterns.

It helps identify:

- Lower-demand hours
- Increasing demand periods
- Peak demand
- Differences between weekdays and weekends

The dashboard identifies 18:00 as the peak demand hour.

26.2 Top 5 Pickup Zones

The Top 5 Pickup Zones visualization identifies locations with the highest taxi demand.

The displayed locations include:

1. Midtown Center
2. Upper East Side South
3. Upper East Side North
4. JFK Airport
5. Times Sq/Theatre District

The chart makes it easy to identify the locations where taxi activity is most concentrated.

26.3 Taxi Demand by Borough

The borough visualization compares taxi demand across:

- Manhattan

- Brooklyn
- Bronx
- EWR
- Staten Island

The visualization shows that Manhattan accounts for a substantial share of the observed taxi demand.

26.4 Actual vs Predicted Taxi Demand

The Actual vs Predicted chart compares the model's predictions against actual demand values across the test period.

The predicted demand generally follows the overall pattern of the actual demand.

This provides a visual representation of how well the model captures changes in taxi demand.

26.5 Model Performance Summary

The dashboard also contains a model performance summary showing:

MAE: 3.63 | RMSE: 11.86 | R²: 96.27%

This allows the machine learning results to be communicated directly through the business intelligence dashboard.

27. DASHBOARD DESIGN AND LAYOUT

The dashboard follows an executive-style layout.

Top Section

The top section contains:

- Dashboard title
- Analysis period
- KPI cards

This provides an immediate summary of the project results.

Middle Section

The middle section contains the main demand analysis:

- Hourly Taxi Demand
- Top 5 Pickup Zones

Lower Section

The lower section contains:

- Taxi Demand by Borough
- Actual vs Predicted Taxi Demand

Bottom Section

The bottom section contains the model performance summary.

This layout allows the dashboard user to move from:

Overall KPIs → Demand Patterns → Geographic Analysis → Prediction Results → Model Performance

in a logical order.

28. KEY BUSINESS INSIGHTS

28.1 Demand Varies by Hour

Taxi demand changes significantly throughout the day.

The analysis shows lower demand during some early hours and increased demand later in the day.

The highest demand hour identified by the dashboard is:

18:00

This highlights the importance of evening demand periods.

28.2 Demand Is Concentrated in Specific Pickup Zones

The Top 5 Pickup Zones visualization shows that demand is concentrated in certain locations.

The highest-demand location is:

Midtown Center

This indicates that location is an important factor when analyzing taxi demand.

28.3 Manhattan Has High Demand Concentration

The borough-level analysis shows that Manhattan has significantly higher taxi demand than the other displayed areas.

This highlights the geographic concentration of taxi activity in the analyzed data.

28.4 Historical Demand Is Important

The feature importance analysis shows that lag_1 is the dominant feature.

Its importance is:

0.894862

The second most important feature is lag_24, with an importance of:

0.073075

This indicates that recent and previous-day demand provide strong predictive information.

28.5 Model Predictions Follow Actual Demand

The Actual vs Predicted visualization shows that predicted demand generally follows the overall pattern of actual demand across the test period.

This supports the strong numerical evaluation results obtained from the model.

28.6 Random Forest Outperformed the Baseline

The Random Forest model achieved:

- MAE = **3.63**
- RMSE = **11.86**
- R^2 = **0.9627**

The baseline achieved:

- MAE = **5.43**
- RMSE = **17.91**
- R^2 = **0.8938**

Therefore, the Random Forest model performed better across all three evaluation metrics.

29. TECHNICAL AND MACHINE LEARNING INSIGHTS

The machine learning stage provided several important technical observations.

29.1 Recent Demand Was the Strongest Predictor

The lag_1 feature was the most important variable in the Random Forest model.

This means that demand in the immediately preceding hour contained substantial information about the following hour's demand.

29.2 Previous-Day Demand Was Also Useful

The lag_24 feature was the second most important feature.

This indicates that demand patterns from the same hour on the previous day contributed useful information.

29.3 Time Features Added Context

The model also included:

- Hour
- Day of week
- Day of month
- Weekend status

These features provided temporal context for the demand prediction task.

29.4 The Model Remained Strong Without lag_1

When lag_1 was removed, the model achieved:

$R^2 = 0.9572$

Although this was lower than the full model's R^2 of 0.9627, the result still demonstrates strong predictive performance.

30. PROJECT OUTCOME

The project successfully produced an end-to-end taxi demand analytics and prediction solution.

Data Analytics Outcome

The project produced:

- Cleaned taxi trip data
- Hourly zone-level demand
- Demand analysis by hour
- Pickup zone analysis
- Borough-level analysis
- Weekday and weekend analysis

Machine Learning Outcome

The project produced:

- Engineered demand features
- Baseline evaluation

- Random Forest Regression model
- Validation results
- Final test results
- Feature importance analysis
- Additional model experiment without lag_1
- Test predictions

Power BI Outcome

The project produced:

- Executive dashboard
- KPI cards
- Hourly demand visualization
- Top 5 pickup zones
- Borough-level demand analysis
- Actual vs Predicted demand visualization
- Model performance summary

The final Random Forest model achieved:

MAE = 3.63

RMSE = 11.86

R² = 96.27%

on the unseen test dataset.

31. LIMITATIONS

Although the model achieved strong performance, the project has several limitations.

31.1 Historical Data Dependence

The model relies heavily on historical demand information.

Unexpected changes in taxi demand may not be captured by the available features.

31.2 Limited Feature Set

The model primarily uses:

- Time information
- Pickup zone

- Historical demand
- Rolling demand

Additional external variables could potentially improve predictions.

31.3 Weather Information Was Not Included

Weather variables were not included in the model.

Rain, snow, temperature, and other weather conditions can affect transportation demand.

31.4 Events Were Not Included

The model does not explicitly include information about:

- Concerts
- Sports events
- Festivals
- Large public gatherings
- Other major events

Such events can cause sudden changes in taxi demand.

31.5 No Real-Time Prediction System

The project focuses on historical analysis and model evaluation.

A production real-time prediction system was not developed as part of this project.

32. FUTURE IMPROVEMENTS

The project could be extended in several ways.

32.1 Weather Features

Weather data could be incorporated to analyze the impact of:

- Rain
- Snow
- Temperature
- Severe weather

on taxi demand.

32.2 Holiday Features

Holiday indicators could be added to capture changes in demand during special dates.

32.3 Event Features

Major events could be incorporated as additional variables.

32.4 Additional Machine Learning Models

Other regression algorithms could be tested and compared with Random Forest.

32.5 Hyperparameter Optimization

Hyperparameter tuning could be performed to determine whether model performance can be further improved.

32.6 Advanced Forecasting Techniques

Future versions could explore specialized time-series forecasting methods.

32.7 Geographic Visualization

Interactive maps could be added to Power BI to provide more detailed geographic demand analysis.

33. CONCLUSION

The **NYC Taxi Demand Prediction** project demonstrates a complete data analytics and machine learning workflow using real-world taxi trip data.

The project began with NYC Yellow Taxi trip data and transformed the raw trip-level information into hourly pickup demand by zone.

Data cleaning and preprocessing prepared the data for analysis, while exploratory analysis helped identify important demand patterns across hours, pickup locations, and boroughs.

Feature engineering introduced time-based and historical demand variables, including previous-hour demand, previous-day demand, and rolling 24-hour demand.

A Random Forest Regression model was then trained using a chronological train-validation-test approach.

The final model achieved:

MAE = 3.63

RMSE = 11.86

$R^2 = 0.9627$

The model therefore explained approximately **96.27% of the variance in the unseen test demand data**.

The Random Forest model also outperformed the baseline model, which achieved an MAE of 5.43, RMSE of 17.91, and R^2 of 0.8938.

Feature importance analysis showed that lag_1, representing previous-hour demand, was the most influential feature, followed by lag_24, representing demand from the same hour on the previous day.

The project was completed with an interactive Power BI executive dashboard that presents the main demand KPIs, hourly demand patterns, top pickup zones, borough-level demand, actual versus predicted demand, and model performance.

Overall, this project demonstrates practical experience across the complete data analytics workflow:

Data Cleaning → Exploratory Data Analysis → Feature Engineering → Machine Learning → Model Evaluation → Data Visualization → Business Intelligence

The project provides a practical example of how raw transportation data can be transformed into meaningful analytical insights and predictive results.

34. PROJECT FILES AND REPOSITORY STRUCTURE

The final project repository is organized as follows:

taxi-demand-prediction/

```
|
|
| — power-bi/
|   | — NYC_Taxi_Demand_Dashboard.pbix
|   | — feature_importance.csv
|   | — hourly_zone_demand.parquet
|   | — ml_predictions.csv
|   | — model_performance.csv
|
|
| — screenshots/
|   | — Dashboard_output.jpg
|
|
| — README.md
|
| — Taxi_Demand_Prediction.ipynb
```

File Descriptions

Taxi_Demand_Prediction.ipynb

Contains the Python-based data analysis, preprocessing, feature engineering, machine learning model development, evaluation, feature importance analysis, and prediction generation.

NYC_Taxi_Demand_Dashboard.pbix

Contains the completed Power BI executive dashboard.

feature_importance.csv

Contains the feature importance results generated from the Random Forest model.

hourly_zone_demand.parquet

Contains the processed hourly zone-level demand dataset used for the analysis and machine learning workflow.

ml_predictions.csv

Contains the actual and predicted demand values generated for the test dataset.

model_performance.csv

Contains the model evaluation results.

Dashboard_output.jpg

Contains the final Power BI dashboard screenshot used as the project preview.

README.md

Contains the concise project overview, methodology, results, insights, workflow, and repository information for the GitHub project.

PROJECT RESULTS SUMMARY

Category	Result
Total Dashboard Trips	~ 3.47M
Unique Pickup Zones	261
Hourly Demand Dataset Rows	194,184
Training Rows	144,072
Validation Rows	18,792
Testing Rows	25,056
Model	Random Forest Regression
Validation MAE	4.17
Validation RMSE	13.45
Validation R ²	0.9402
Final Test MAE	3.63
Final Test RMSE	11.86

Final Test R ²	0.9627 / 96.27%
Baseline MAE	5.43
Baseline RMSE	17.91
Baseline R ²	0.8938
Peak Demand Hour	18:00
Top Pickup Zone	Midtown Center
Most Important Feature	lag_1
lag_1 Importance	0.894862
lag_24 Importance	0.073075
R ² Without lag_1	0.9572 / 95.72%